

Article Critique 1

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For my article critique, I chose to review the peer-reviewed publication entitled “Mandatory Hepatology Education for Internal medicine Residents: Long-Term Effects and Implications for Workforce Needs” by Mikolajczyk et al. (2021). This single-center quantitative study conducted between 2014-2018 involved the creation of a mandatory two-week hepatology inpatient rotation for internal medicine residents along with a complementary didactic curriculum to improve hepatology foundational knowledge and potentially stimulate specialty career interest. Data was also compared against a historical study involving a similar cohort at the same institution however the original study did not assess long-term follow-up unlike the current study design. The use of long-term comparison across the same cohort is a strength of the study and builds on the author’s prior work and research in this domain. Having historical data as a comparison point (albeit different cohort, but same institution) builds upon this strength even further. However, the study design incorporating only a 2-week mandatory inpatient rotation and no ambulatory or outpatient exposure does provide a one-sided clinical experience and misses out on the rather substantial outpatient predominant nature of the field. Hence, evaluating for changes in specialty interest based solely on inpatient experiences where the specialty is predominantly outpatient based does detract from the study design and a weakness overall.

The total resident population appeared to be recruited for this study given the mandatory nature of the study design and hence sampling was not required. Nevertheless, a sample size calculation could have been carried out to see if extending the study by 1-2 years would be necessary in order to appropriately power the study to support their objectives. Had this data been included, that would have strengthened the intervention overall and put the outcomes in better perspective.

Supplement 1 provides necessary and pertinent details related to the hepatology rotation and implemented curriculum. One of the educational interventions was “participation in patient care, with careful attention to ensure residents independently worked through each of the four steps in Kolbe experiential learning theory (e.g. concrete experience, reflective observation, abstract conceptualization, and active experimentation) in each patient encounter.” While the specific reference to educational theory is appreciated, one needs to extrapolate how the further listed resident responsibilities more specifically reflect each of the learning theory steps. Details are not provided whether supervising faculty had experience or familiarity with Kolbe experiential learning theory, nor whether learners that were performing below expectations were accounted for in curricular design which are also areas for improvement.

Regarding assessment tools, Likert-type questions were developed to evaluate topic comfort and perceived topical knowledge and used in combination with standardized board-review questions from an established reference (Medical Knowledge Self-Assessment Program). I appreciated the detail provided by the authors that the Likert-type questions developed and used did not have prior validity evidence as this is important for

the readers to recognize when critically evaluating the study and to consider if implementation of a similar mandatory hepatology education program is planned at other institutions. The combination of these identical Likert-type questions along with board-review questions from a reputable, standardized question bank for both pre-residency and post-residency assessments to objectively judge improvement in clinical knowledge after the intervention is a strength of the study.

Regarding statistical analysis, I appreciated that the authors clearly stated the software used (RStudio) and provided a clear description of how missing or partial data was interpreted in the analyses with students that missed the demographic questions but otherwise answered the remainder of the questions included in analysis (but those with only partial assessments were excluded). Wilcoxon rank sum tests were utilized for statistical analysis of the Likert-type questions which are the appropriate tests for non-parametric data and for comparing medians across independent samples. Given the study design, the output would not be normally distributed, and these tests would be appropriate. I do feel that there is a small bit of confusion over the stated non-parametric tests used as Wilcoxon rank sum tests and Mann-Whitney U-tests seem to be the two different names for the same test. Further clarification would be appreciated not only for reproducibility but to also better interpret the statistical analysis. T-tests were also utilized for comparison across means for the standardized board review questions which would be the appropriate testing for this group. Beyond the small weakness described above, the clear presentation of data in the Results section was effectively done through figures and tables which is an overall strength.

Finally, the discussion and conclusion sections are overall well written and appropriately review some of the limitations of the study. The potential for intervention/exposure bias was addressed and explained which is an appreciated detail. Surprisingly specialty interest in gastroenterology and hepatology declined in the intervention cohort which seems paradoxical given the increased dedicated education and patient exposure during the 2-week mandatory clinical rotation. This is only briefly addressed in the discussion section however though I do agree that only having an inpatient exposure to these specialty fields without ambulatory clinical experiences could impact future career interest. Future areas of investigation are also explored which is a strength of the article, as this is a key detail to help guide future research in this domain and potentially spark research ideas among readers.

Reference:

Mikolajczyk, A. E., Zilberstein, N., McConville, J. F., Pan, A., Aronsohn, A. I., Te, H. S., Reddy, G., Paul, S., Pillai, A., Charlton, M., & Farnan, J. M. (2021). Mandatory Hepatology Education for Internal Medicine Residents: Long-Term Effects and Implications for Workforce Needs. *Hepatology communications*, 5(11), 1953–1963.
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